

Homotopy Pushouts and Singular Chains

Algebraic Topology I | Fall 2026 | October 2026

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Lecture 7 established the topological gluing results and the algebraic double mapping cylinder for singular chains. We now develop the mapping complexes and comparison maps that make that construction apply to arbitrary complexes. The topological definitions, cofibration lemma, and examples remain in Lecture 7, §1; its Definition 2.4 fixes the cone and sign conventions used below.

1 Mapping Complexes and Chain Homotopies

The tensor product and mapping complex express chain homotopies through an interval complex. We use the homological grading and the category $\text{Ch}(\mathbb{Z})$ of Lecture 2. For a complex A_* , write d_A for its differential.

1.1 The Tensor–Mor Adjunction

The mapping complex retains graded maps as well as chain maps. Its differential records which graded maps commute with the differentials of the two complexes.

Definition 1.1. For complexes A_*, K_*, B_* of abelian groups, define the tensor product by

$$(A \otimes K)_n = \bigoplus_{p+q=n} A_p \otimes_{\mathbb{Z}} K_q, \quad d(a \otimes k) = d_A a \otimes k + (-1)^p a \otimes d_K k \quad (a \in A_p).$$

Define the *mapping complex* by

$$\underline{\text{Mor}}(A, B)_r = \prod_{n \in \mathbb{Z}} \text{Mor}_{\mathbb{Z}}(A_n, B_{n+r}), \quad \partial \phi = d_B \phi - (-1)^r \phi d_A.$$

Thus ∂ lowers degree by one. Its degree-zero cycles are chain maps, and a degree-one element s with $\partial s = f_1 - f_0$ is a chain homotopy from f_0 to f_1 .

Composition of graded maps gives a chain map

$$\underline{\text{Mor}}(B, C) \otimes \underline{\text{Mor}}(A, B) \rightarrow \underline{\text{Mor}}(A, C).$$

Indeed, for ψ of degree s , $\partial(\psi \phi) = (\partial \psi) \phi + (-1)^s \psi(\partial \phi)$. Composition is associative, and the identity maps are degree-zero cycles. Hence the morphisms between complexes themselves form complexes, compatibly with composition.

Proposition 1.2. For complexes of abelian groups, there is a natural isomorphism of chain complexes

$$\underline{\text{Mor}}(A \otimes K, B) \cong \underline{\text{Mor}}(A, \underline{\text{Mor}}(K, B)).$$

It sends a homogeneous map F to the map $F^\#$ defined by $F^\#(a)(k) = F(a \otimes k)$.

Proof. Degreewise, the isomorphism is the tensor–Mor adjunction for abelian groups, with the direct sums in the tensor product becoming products of morphism groups. For F of degree r and $a \in A_p$, the two differentials agree since

$$\begin{aligned} (\partial F)^\#(a)(k) &= d_B F(a \otimes k) - (-1)^r F(d_A a \otimes k) \\ &\quad - (-1)^{r+p} F(a \otimes d_K k) \\ &= (\partial F^\#)(a)(k). \end{aligned}$$

□

For fixed K , this says concretely that forming $A \otimes K$ is left adjoint to forming $\underline{\text{Mor}}(K, B)$, including at the level of mapping complexes.

1.2 The Interval and Derived Morphisms

Use the interval complex $\mathbb{Z}[\Delta_{\text{nd}}^1]$ and its generators e_0, e_1, e_{01} from Lecture 7, Definition 2.4. The tensor differential explains the sign in the degree-one component of its cylinder homotopy. For chain maps $f_0, f_1 : A_\bullet \rightarrow B_\bullet$, a chain homotopy from f_0 to f_1 is equivalently a chain map

$$h : A_\bullet \otimes \mathbb{Z}[\Delta_{\text{nd}}^1] \rightarrow B_\bullet$$

whose restrictions to the endpoints are f_0, f_1 . Indeed, for $a \in A_n$, a degree-one map $s : A_\bullet \rightarrow B_{\bullet+1}$ determines the graded map h by

$$h(a \otimes e_0) = f_0(a), \quad h(a \otimes e_1) = f_1(a), \quad h(a \otimes e_{01}) = (-1)^n s(a).$$

With the tensor differential of Definition 1.1, the chain-map identity on $a \otimes e_{01}$ is

$$(-1)^n d_B s(a) = (-1)^{n-1} s(d_A a) + (-1)^n (f_1(a) - f_0(a)),$$

or $d_B s + s d_A = f_1 - f_0$. The identities on the two vertices say that f_0, f_1 are chain maps.

Remark 1.3. For an additive category $\mathcal{A}$, the mapping complex is defined in the same way using $\text{Mor}_{\mathcal{A}}$, with the products taken in abelian groups. Its zeroth homology gives the morphisms in the homotopy category:

$$\text{Mor}_{K(\mathcal{A})}(A_\bullet, B_\bullet) = H_0 \underline{\text{Mor}}_{\mathcal{A}}(A_\bullet, B_\bullet).$$

Indeed, cycles are chain maps and boundaries are differences of chain-homotopic maps. For an abelian category $\mathcal{A}$, use the homotopy and derived categories $K(\mathcal{A})$ and $D(\mathcal{A})$ of Lecture 2. Thus derived morphisms are not in general computed by the unchanged mapping complex. For modules, if $B_n = 0$ for $n \gg 0$, choose a quasi-isomorphism $B_\bullet \rightarrow I_\bullet$ with $I_\bullet$ bounded above and termwise injective. Then

$$\text{Mor}_{D(\mathcal{A})}(A_\bullet, B_\bullet) \cong H_0 \underline{\text{Mor}}_{\mathcal{A}}(A_\bullet, I_\bullet).$$

Here $\mathcal{A}$ is the module category. This is the bounded-above injective-resolution computation in homological grading; see the Stacks Project, Lemma 13.18.2 and Lemma 13.18.8, written there in cohomological grading.

2 Comparison Maps for Homotopy Pushouts

A square with a chosen chain homotopy induces a comparison from the cone construction of Lecture 7. The degree-one component of the homotopy supplies the third term of that comparison.

2.1 The Comparison Map and Its Signs

The following square is the algebraic setting for the comparison. The endpoint maps and the chosen homotopy are retained as part of its data.

Definition 2.1. A homotopy-commutative square of complexes consists of chain maps

$$\begin{array}{ccc} A_{\bullet} & \xrightarrow{f} & B_{\bullet} \\ g \downarrow & & \downarrow i \\ C_{\bullet} & \xrightarrow{j} & D_{\bullet} \end{array}$$

and a specified chain map

$$h : A_{\bullet} \otimes \mathbb{Z}[\Delta_{\text{nd}}^1] \rightarrow D_{\bullet}, \quad h(a \otimes e_0) = if(a), \quad h(a \otimes e_1) = jg(a).$$

For $a \in A_n$, define its degree-one component by $s(a) = (-1)^n h(a \otimes e_{01})$. The chain-map identity for h is equivalent to

$$d_D s + s d_A = jg - if. \quad (1)$$

Thus h is a homotopy from if to jg , represented by the degree-one map s .

For the span $B_{\bullet} \xleftarrow{f} A_{\bullet} \xrightarrow{g} C_{\bullet}$, use the cone construction of Lecture 7, Definition 2.4, with $A_{\bullet}, B_{\bullet}, C_{\bullet}$ in place of the three singular chain complexes. Denote the resulting complex by $Z_{\bullet}$ and its structure maps by i_Z, j_Z, h_Z . Write s_Z for the degree-one component of h_Z . A degree- n element of $Z_{\bullet}$ is written (b, c, a) , where $b \in B_n, c \in C_n$, and $a \in A_{n-1}$. The construction applies to arbitrary chain maps because its differential and homotopy identities use only the chain-map identities for f and g .

Proposition 2.2. For a square (f, g, i, j, h) as in Definition 2.1, there is a chain map

$$\Phi_h : Z_{\bullet} \rightarrow D_{\bullet}, \quad \Phi_h(b, c, a) = -i(b) + j(c) + s(a). \quad (2)$$

Its composites with the two structure maps i_Z, j_Z are i and j .

Proof. The chain-map condition is the homotopy identity (1). For $(b, c, a) \in Z_n$, expand the two sides:

$$\begin{aligned} \Phi_h d_Z(b, c, a) &= -id_B b + jd_C c + (jg - if)(a) - sd_A a \\ &= -id_B b + jd_C c + d_D s(a) && \text{by (1)} \\ &= d_D(-i(b) + j(c) + s(a)) && \text{since } i, j \text{ are chain maps} \\ &= d_D \Phi_h(b, c, a). \end{aligned}$$

Finally, $\Phi_h(-b, 0, 0) = i(b)$ and $\Phi_h(0, c, 0) = j(c)$, giving the required composites. $\square$

Definition 2.3. A square (f, g, i, j, h) of complexes is a *homotopy pushout square* if its comparison map Φ_h in Proposition 2.2 is a quasi-isomorphism.

This is a condition in the homotopy theory of complexes with quasi-isomorphisms as equivalences. With the structure maps i_Z, j_Z, h_Z , the canonical complex $Z_{\bullet}$ is a homotopy pushout: its comparison map to itself is the identity. The square with lower-right complex $D_{\bullet}$ is a homotopy pushout when Φ_h is a quasi-isomorphism. The sign in its first structure map makes the comparison map restrict to the given maps i and j .

2.2 The Long Exact Homology Sequence

The shifted common complex occurs as a quotient of the double mapping cylinder. This quotient produces the connecting map, including its sign.

Apply the mapping-cone long exact sequence of Lecture 2 to $(f, g) : A_\bullet \rightarrow B_\bullet \oplus C_\bullet$. Let $k : B_\bullet \oplus C_\bullet \rightarrow Z_\bullet$ be the summand inclusion and let $p : Z_\bullet \rightarrow A_\bullet[1]$ be the projection onto the shifted summand.

For a cycle $a \in A_{n-1}$, the lift $(0, 0, a) \in Z_n$ has boundary $(f(a), g(a), 0)$. Hence the boundary homomorphism out of $H_n(A[1])$ is (f_*, g_*) . The projection sends $[(b, c, a)]$ to $[a]$ under the shift identification. We take the negative of this projection on homology, so that the convention agrees with the Mayer-Vietoris sequence in Lecture 7:

$$\delta_Z[(b, c, a)] = -[a]. \quad (3)$$

Negating a homomorphism leaves its kernel and image unchanged, so this convention preserves exactness. The resulting class is independent of the chosen cycle representative because p is a chain map.

Theorem 2.4. For a homotopy pushout square (f, g, i, j, h) of complexes, there is a long exact sequence

$$\begin{aligned} \cdots \rightarrow H_n(A) &\xrightarrow{(f_*, g_*)} H_n(B) \oplus H_n(C) \xrightarrow{j_* - i_*} H_n(D) \\ &\xrightarrow{\delta_D} H_{n-1}(A) \xrightarrow{(f_*, g_*)} H_{n-1}(B) \oplus H_{n-1}(C) \rightarrow \cdots \end{aligned} \quad (4)$$

Its connecting map is $\delta_D = \delta_Z(\Phi_{h,*})^{-1}$.

Proof. By Definition 2.3, the homomorphism $\Phi_{h,*} : H_n(Z) \rightarrow H_n(D)$ is an isomorphism in every degree. Transport the mapping-cone exact sequence along these isomorphisms, using the convention (3). Formula (2) gives

$$\Phi_h k(b, c) = j(c) - i(b),$$

so the map into $H_n(D)$ is $j_* - i_*$. The transported connecting map is the one stated, proving (4). $\square$

A cycle $z \in D_n$ can therefore be handled as follows. Choose a cycle $(b, c, a) \in Z_n$ whose image under Φ_h is homologous to z . The cycle conditions and the image are

$$\begin{aligned} d_A a &= 0, & d_B b &= -f(a), & d_C c &= -g(a), \\ z &\sim -i(b) + j(c) + s(a). \end{aligned}$$

Then $\delta_D[z] = -[a]$. The common chain a records the boundary that has to be matched between the two pieces. The preceding projection argument proves that this class is independent of the representative in Z_n .

For the mapping-cone specialization $C_\bullet = 0$, the map from $H_n(B)$ is induced by the negative of the usual summand inclusion. This is the same sign convention as in Lecture 7; the mapping-cone and suspension calculations themselves are given there and in Lecture 2.

References. Lecture 2, for mapping cones, homotopy categories, and derived categories. Lecture 7, §1 and Definition 2.4, for topological gluing and the algebraic double mapping cylinder. The Stacks Project, Lemma 13.18.2 and Lemma 13.18.8, for injective resolutions and derived morphisms.